Effect

New jobs

The production, distribution and planting of seed balls creates new jobs, contributing to an improved economy and standard of living

Impact on the environment

It helps with forest restoration, recovery, and growth. Based on estimates from the production of the balloons over the next 25 years and the impact on forests, the project could reduce the Earth's average temperature by 0.006 degrees Celsius by 2050.

Sustainable Future

promotes the development of a green economy and attracts investment in projects aimed at a green future, motivating people to participate in environmental initiatives

Improving air quality

Urban trees can reduce fine particulate matter (PM2.5) levels by 7–24%, removing approximately 711,000 tons of pollutants from urban air.

Development Roadmap

Mid-2026

Developing and completing research and product development, obtaining feedback from botanical experts, obtaining copyright and patent, and assessing actual revenues and expenses

2026 - early 2027

Search for beneficiaries and investors Targeted advertising for production Distribution of the product on the domestic market, obtaining subsidies for the production of the product

2027-2029

Use of the product for land reclamation after mining and metallurgical activities Adaptation of the composition to the climatic conditions of the regions of Kazakhstan



enactus KAZAKHSTAN

NIS Karaganda 2

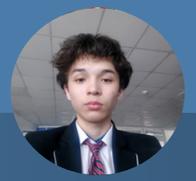
PROGRESS REPORT 2025 — 2026

NIS Karaganda Team No. 2

11th grade



Amanzhol Dias
Engineer, Front-End Developer



Mukhtarkhan Batyrkhan
Lead Economist



Kulkenova Maryam,
PR manager, SMM



Timofey Shchannikov,
Back-End Developer



Shtro Artur, team captain



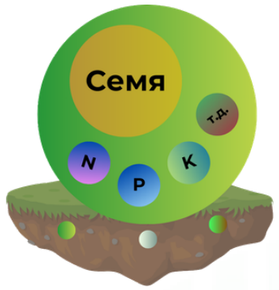
Abeuov Yerassyl
designer, construction engineer

Problem

Since 1990, 420 million hectares of forest have been lost worldwide due to deforestation.

Global deforestation accounts for 20% of greenhouse gas emissions

From 2015 to 2020, forest cover loss amounted to 10 million hectares annually.



Solution

Green Orbs

Balls containing seeds of fast-growing tree species, nutrients and growth stimulants.

AI drones distribute the balls across the area, and the plant begins to grow independently thanks to a supply of nutrients that is sufficient for growth for the first 2-3 years, after which the tree independently ensures its own vital processes.

The purpose and essence of the project

The creation of a profitable business project, GreenOrbs, which will enable the efficient, rapid, large-scale, and convenient greening of large areas in various climatic conditions, becoming a successful ESG solution for companies.

Green Orbs is Kazakhstan's first AI-powered bioengineered precision drone seeding system for arid climates with pest control.

This innovative approach allows you to green large areas in just a few hours, ESPECIALLY in hard-to-reach places. It's much more efficient and cost-effective than manual labor.

drone

EYES NDVI (Normalized Vegetation Index)

Mapping allows for the identification of "microsites," specific zones in the soil that promote moisture retention. Rather than randomly dropping seeds, the drone targets these viable zones to maximize survival.

BRAIN The software generates a flight path

HANDS The low-speed dosing auger rotates over time, releasing exactly one or two balls

Ball

Bioactive matrix (60%)

50% bentonite clay (binder and structural integrity)

25% vermicompost (nutrient)

15% biochar (filters toxins and masks odors)

5% superabsorbent polymer (hydrogel for water retention)

<1% capsaicin (pepper extract for rodent repellent).

Nutrients (20%)

The composition is adapted to the specific terrain. The 60/20/20 composition ensures the highest probability of tree planting.

Seed (20%)

10%
support trees (elm/poplar)

40%
shrubs (saxaul/tamarisk)

50%
groundcover plants (grass/clover)

Results of the team's work

>1850 total hours of work on the project

3 finals: Astana Innovation Forum, BECO Hackathon, ATAMEKEN StartUp

conducted

Observations of tree growth from different types of seeds



GreenOrbs' best option

conducted

Experiments

At the Department of Botany at the Buketov KarSU, the optimal composition of the balls was determined and the most optimal variant was identified.

held

Fertilizer research

- **Growth-regulating hormones (gibberellins and rooting hormones, which stimulate root formation and initial growth)**
- Essential macro- and microelements: (N, P, K, Mg, Fe)



GreenOrbs composition

developed

Website prototype

acting as a channel of interaction with clients and as a way to promote the project

planted

More than 80 samples

on the territory of the NIS EMN in Karaganda and in the Triathlon Park as part of the park landscaping program with the support of the Karaganda City Akimat



developed

Improved financial model

B2B, B2C, B2G analysis, market analysis, competitor analysis, potential partner analysis, detailed analysis of initial and annual expenses, product payback (revenue) predictions

a full audit was conducted

Available Market Analysis

TAM: The global reforestation market is estimated to be around $50–55 billion by 2025–26, with a projected increase of 8–10% by 2030.
SOM: The state program "Zhasyl Kazakhstan" under the supervision of the Ministry of Ecology and Natural Resources of the Republic of Kazakhstan, where investment volume exceeds 1.4 trillion tenge. Major subsoil users have spent over 12.5 billion tenge on soil reclamation.
SAM: interaction with companies using drones (57 drones in the country's agricultural sector by 2025), and direct sales (cooperation with akimats, environmental organizations: AEOK), participation in tenders

defined

Strategic partners:

Seed/Fertilizer and Drone Suppliers – AlemAgro, KazAzot, KazPhosphorus and the Skyworker branch in Karaganda

Research institutes – Kazakh Research Institute of Agriculture and Plant Growing, Faculty of Botany, Karaganda State University named after Buketov